

### Program 7

The following program accepts a number from the user. If the user enters an even number, it stops execution.

```
#include<iostream.h>
int main()
{
    int x;
    char check=' y' ;
    while(check==' y' )
    {
        cout<<"Execution will stop if you enter even
number";
        cout<<"\nEnter a number=>";
        cin>>x;
        if(x%2==0)
        {
            cout<<"\nyou have entered an even number";
            check=' n' ;
        }
    }
}
```

Here, the program first creates an integer variable 'x' and a character variable 'check'. The character variable is then initialized by 'y'. If the condition of 'while' holds true, the control enters the loop. Here, it prompts the user to enter a number. If the user enters an odd number, then the 'if' condition holds false. The value of 'check' will remain 'y'. Therefore, the condition of the loop holds true and the body of the loop is executed again. This process continues till the user enters an even number. If the user enters an even number, the 'if' condition holds true and its body is executed which prints a message. The value of check becomes 'n'. Finally, the condition of the loop is checked, and the execution of the program ends once the condition is false.